

ISRM v1.0 — Final Internal Validation Addendum

Edge-case, adversarial and concurrency validation

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1. Purpose

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This addendum records the final internal stress-testing performed before the ISRM package is presented to DIN for technical consideration. It supplements the reference-implementation validation report and focuses on boundary conditions that can invalidate an otherwise plausible reference architecture: races, stale authorization, replay, cyclic delegation, uncertain execution state, conflicting authority, concurrent adaptation, compensation and incomplete state.

2. Test outcome

Metric	Result
Edge-case test executions	60
PASS	60
FAIL	0
ERROR	0
Property-test iterations per invariant family	50,000
Observed invariant violations	0

3. Edge-condition families

- **Authority races and revocation:** Authorization is re-evaluated where relevant; stale or revoked authority does not silently survive into execution.
- **Replay and duplicate delivery:** Repeated messages/actions are constrained by episode identity, authorization and idempotency semantics.
- **TOCTOU / stale decisions:** A decision based on state that materially changed before execution is not treated as automatically valid.
- **Partial and unknown execution:** Attempted execution, confirmed effect, partial effect and unknown outcome remain distinguishable.
- **Compensation and late success:** A late success after compensating action is represented as a new observed state rather than overwritten history.

- **Cyclic and nested delegation:** Recursive composition terminates or rejects cyclic delegation; delegated authority remains bounded.
- **Conflicting concurrent episodes:** Concurrent goals competing for the same resource require arbitration rather than accidental last-writer-wins semantics.
- **Adaptation during operation:** Behavior-changing adaptation is versioned and cannot silently alter the authority/safety basis of an in-flight episode.
- **Uncertainty and incomplete state:** Material uncertainty is preserved and may force defer, request-information or escalation.
- **Trust-boundary attacks:** Untrusted instructions, identity changes and provenance loss do not automatically become authorized intent.

4. Property-based invariant tests

- Authority monotonicity: delegated effective authority never exceeds the authority granted by the delegator unless a separate explicit authority event exists.
- Delegation intersection: effective delegated capability is bounded by the intersection of requested capability and delegable authority.
- Uncertainty preservation: uncertainty material to a consequential decision cannot disappear merely because an object crosses a layer boundary.
- Candidate/authorization separation: generation of a Candidate Action never alone creates an Authorized Action.
- Idempotent execution semantics: replay of an operation declared idempotent cannot be interpreted as a new independent authorization.

5. Interpretation

The result supports internal consistency and implementability of the current ISRM semantics across the tested cases. It does not prove completeness, novelty, legal compliance, safety for a particular application, or suitability as an International Standard. Independent implementation, standards-committee review and clause-level comparison with applicable standards remain required.

6. Submission freeze

For the DIN approach, the architecture is frozen at ISRM v1.0. Any subsequent substantive semantic change shall receive a new version identifier and shall not be represented as the same tested release. Editorial corrections that do not change normative meaning may be tracked separately.

7. Final internal gate

Result: PASS FOR DIN TECHNICAL SUBMISSION / DISCUSSION.

ISO/IEC submission is not a direct individual filing: a New Work Item Proposal to an existing ISO committee is submitted by an eligible proposer such as an ISO member body, committee secretariat, liaison organization, TMB/advisory group or the Secretary-General. For a German originating proposal, the intended route is therefore DIN/NA 043-01-42 GA → German support/proposal → ISO/IEC JTC 1/SC 42.